

Upper Jurassic and Lowermost Cretaceous Boreal Radiolarian Assemblages of Russia and Their Paleobiogeographic and Stratigraphic Significance: a Review

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Abstract—Boreal radiolarians of the Upper Jurassic (Oxfordian, Kimmeridgian and Volgian) and of the lower part of Lower Cretaceous (Ryazanian and Valanginian) of Russia are known from the territories of the Eastern European Craton, Timan-Pechora Basin, Western Siberian Basin and northern periphery of the Siberian Craton. Radiolarian assemblages of this stratigraphic interval are characterized by dominance of high-latitude taxa (genera *Parvicingula* and *Praeparvicingula* of the family Parvicingulidae) and common presence of the typically boreal family Echinocampidae (genera *Echinocampe*, *Nordvikella* and *Arctocapsula*). The genus *Pantanellium* is present in the Kimmeridgian of Eastern European Craton and Timan-Pechora Basin. This genus is represented by the *P. meraceibaense* a group of species, that can spread to southern boreal areas. Previous reports of the presence of the subfamily Vallupinae, which is typical of low paleolatitudes, cannot be confirmed. Radiolarian assemblages of the Boreal Upper Jurassic and lower part of Lower Cretaceous show considerable taxonomic changes through the succession and can therefore be used in stratigraphy.

Keywords: Parvicingulidae, Echinocampidae, Pantanelliidae, Eastern European Craton, Western Siberia, Arctic Siberia

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INTRODUCTION

Upper Jurassic and Lower Cretaceous Boreal radiolarians have been known in Russia since the 1930s. The earliest publication on these radiolarians was by Khudyaev (1931), discussing radiolarians in the Neocomian (most likely, Ryazanian-Valanginian) and Kimmeridgian phosphorites of the Sysola River basin (Komi Republic). Later, Khabakov (1937) described the radiolarian assemblages of the Upper Jurassic (Kimmeridgian and Volgian Stage) and Lower Cretaceous (Ryazanian Stage or Valanginian) from phosphorites of the upper reaches of the Kama and Vyatka rivers. At that time, radiolarians from dense rocks were studied exclusively in thin sections, and taxa (genera and species) cannot always be identified from the images available in these works, or only with some degree of conventionality. Nevertheless, these data are of interest and, if possible, will be used in this publication.

The 3D forms of boreal radiolarians in Russia were first studied by Kozlova (1971, 1973) from the Kimmeridgian deposits of the Timan-Pechora Plate. From that time, an increasingly intense study of radiolarians has begun. In recent decades, at least two dozen Russian-language works have been published, covering the interval from the Oxfordian to the Valanginian. It

is noteworthy that the Upper Jurassic and Lower Cretaceous radiolarians from the Boreal regions have been studied in detail by researchers in this country, while publications in this field, as well as records of Boreal radiolarians of this age interval outside Russia, are few: one can mention studies of radiolarians of the Upper Jurassic – Lowermost Cretaceous of the North Sea (Dyer and Copestake, 1989) and the Upper Jurassic (Volgian Stage) of Spitsbergen (Nakrem and Kiessling, 2012).

All publications to date cover the interval from the Oxfordian to the Valanginian and are confined to four large regions: the East European Craton, the Timan-Pechora Plate, the West Siberian Plate and the northern framing of the Siberian Craton (Figs. 1, 2). The studied sections are generally well characterized by fossils (ammonites, bivalves, foraminifers, etc.), providing a reliable stratigraphic reference of radiolarian assemblages. In most cases, the studied radiolarians are satisfactorily and well preserved, allowing a detailed study of the morphology and an adequate description of new taxa. The available material makes it possible to assess the stratigraphic and paleobiogeographic significance of the Boreal radiolarian assemblages of the Upper Jurassic and the Lowermost Cretaceous of Russia.

Jurassic			Cretaceous		System
Oxfordian	Kimmeridgian	Volgian	Ryazanian	Valanginian	Stage
Bragin and Kiselev, 2013	Bragin, 1997	Kozlova, 1994			East European Craton
		Vishnevskaya, 1998			
		Vishnevskaya, 2001			
		Vishnevskaya and Baraboshkin, 2001			
		Vishnevskaya and Murchey, 2002			
Palechek and Ustinova, 2020					Timan—Pechora Plate
Bragin et al., 2024	Kozlova, 1971, 1973	Kozlova, 1994			
	Vishnevskaya, 1998				
	Vishnevskaya, 2001				
	Vishnevskaya and Murchey, 2002				
		Vishnevskaya, Kozlova, 2012			West Siberian Plate
		Kozlova, 1983			
		Kozlova, 1994			
		Lipnitskaya, 2006			
		Amon, 2011			
		Vishnevskaya et al., 2020			
		Amon et al., 2021			Northern margins of the Siberian Platform
		Bragin, 2009			
		Bragin, 2011			
		Vishnevskaya et al., 2014			

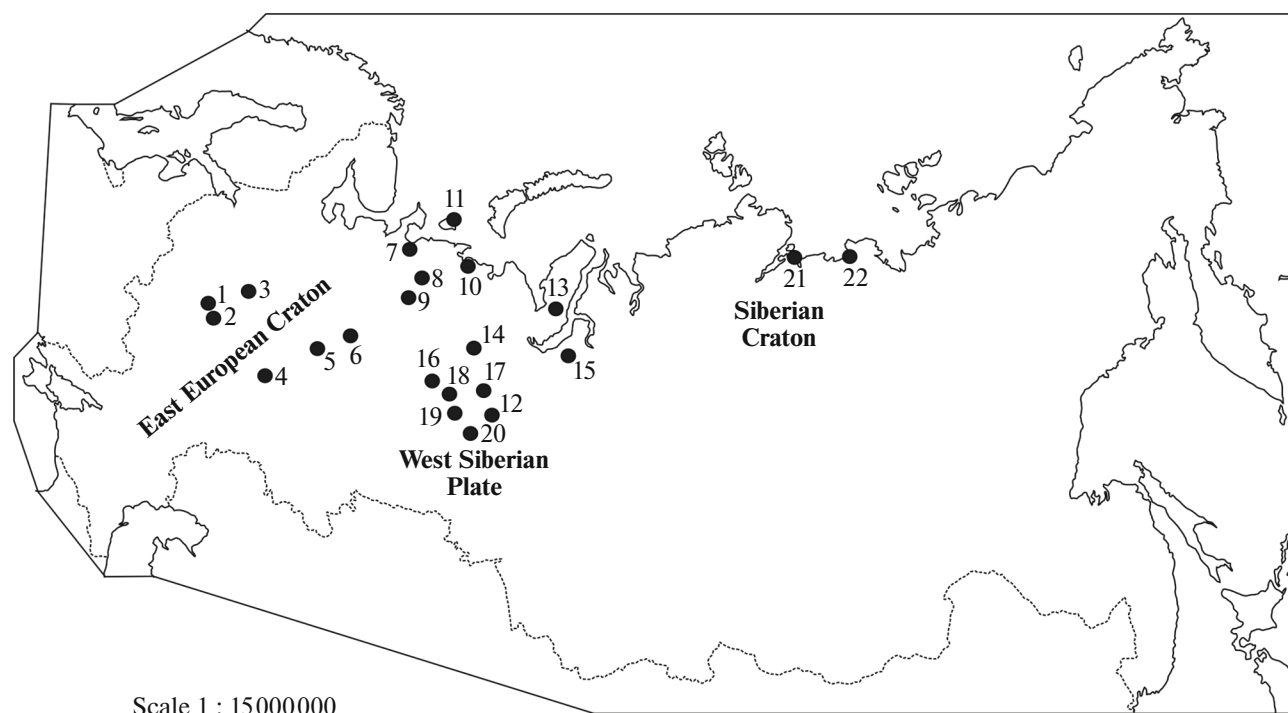


Fig. 2. The most important localities of Oxfordian–Valanginian Boreal radiolarians in Russia. (1) Moscow, Kolomenskoye, Kimmeridgian (Bragin, 1997); Krylatskoye, Oxfordian (Ustinova et al., 2014); Vorobyovy Gory, Oxfordian and Kimmeridgian (Palechek and Ustinova, 2020); (2) Rybaki, Oxfordian (Bragin et al., 2024); (3) Ioda and Cheryomukha rivers, Oxfordian and Kimmeridgian (Bragin and Kiselev, 2013); (4) Gorodishchi, Volgian and Ryazanian stages (Kozlova, 1994; Vishnevskaya, 1998, 2001; Vishnevskaya and Baraboshkin, 2001; Vishnevskaya and Murchey, 2002); (5) upper reaches of the Vyatka and Kama rivers, Kimmeridgian, Volgian and Ryazanian stages, Valanginian (Khabakov, 1937); (6) Koygorodok and Kozhim, Kimmeridgian, Ryazanian stage, Valanginian (Khudyayev, 1931); (7) Pesha River, Volgian Stage (Vishnevskaya, 2017); (8) Pizhma River, Kimmeridgian (Kozlova, 1971, 1994); (9) Ukhta, Kimmeridgian (Vishnevskaya, 1998; Vishnevskaya and Pralnikova, 1999; Vishnevskaya, 2001; Vishnevskaya and Murchey, 2002); (10) Naryan-Mar borehole, Kimmeridgian and Volgian stage (Kozlova, 1994; Vishnevskaya and Kozlova, 2012); (11) Kolguev Island, Kimmeridgian, Volgian and Ryazanian stages (Kozlova, 1994; Vishnevskaya and Pralnikova, 1999; Vishnevskaya and Kozlova, 2012); (12) Upper Salym, Volgian, Ryazanian and Valanginian (Vishnevskaya and Kozlova, 2012; Vishnevskaya et al., 2020; Amon et al., 2021); (13) Yamal, Volgian (Vishnevskaya et al., 2020; Amon et al., 2021, 2022); (14) Molodezhnoye, Volgian, Ryazanian and Valanginian (Vishnevskaya et al., 2020; Amon et al., 2021); (15) Gubkinskoye, Volgian, Ryazanian and Valanginian (Vishnevskaya et al., 2020; Amon et al., 2021); (16) Nizhne-Yanlotskoe (Vishnevskaya et al., 2020); (17) Aprelskoe; (18) Pravdinskoe, Volgian and Ryazanian stages (Panchenko et al., 2015; Vishnevskaya et al., 2020); (19) Radonezhskoe, Volgian Stage, Ryazanian Stage and Valanginian (Vishnevskaya et al., 2020); (20) Malobalykskoe, Volgian Stage, Ryazanian Stage and Valanginian (Vishnevskaya et al., 2020); (21) Nordvik, Volgian and Ryazanian stages (Bragin, 2009, 2011); (22) Ystannakh-Khocho, Ryazanian Stage (Vishnevskaya et al., 2014).

while one of the groups of species of the genus *Pantanelium* Pessagno (*P. meraceibaense* Pessagno et MacLeod) can spread to southern boreal waters (Pessagno et al., 1987). These ideas were confirmed by many works on the Upper Jurassic and Lower Cretaceous of Russia (Bragin, 1997, 2009, 2011; Vishnevskaya, 1998, 2001; Bragin and Bragina, 2018; Vishnevskaya and Murchey, 2002; Vishnevskaya et al., 2020; Amon et al., 2021, 2022).

OXFORDIAN

Oxfordian radiolarian assemblages are relatively rare. Until now, several finds have been made only on the East European Craton, within the Moscow Syncline. The earliest of the known Jurassic boreal radiolarian assemblages was studied in sections of three

boreholes on Vorobyovy Gory (Moscow) from the deposits of the Podosinkovskaya Subformation (upper subformation of the Chulkovskaya Formation) (Palechek and Ustinova, 2020). The subformation is dated by ammonites as late Callovian–early Oxfordian, while the foraminiferal assemblage from the boreholes is determined as, most likely, early Oxfordian (Palechek and Ustinova, 2020). The Upper Callovian–Lower Oxfordian radiolarian assemblage is not rich and is represented by several species, including representatives of the genus *Parvicingula* Pessagno. The species *P. elegans* Pessagno et Whalen, characteristic of the Boreal Upper Jurassic, has been identified here (Vishnevskaya, 2001). Several more specimens of *Parvicingula* Pessagno have been identified in open nomenclature or to genus level (it is possible that they also belong to *P. elegans* Pessagno et Whalen, since the

apical horn in these specimens is most likely absent due to incomplete preservation). Representatives of the genera *Praeconocaryomma* Pessagno and *Higumastra* Baumgartner, which have a worldwide distribution, are also present. Finally, of particular interest is the presence of the species *Mizukidella mokaensis* O'Dogherty, Goričan et Gawlick, which is common in the Upper Jurassic of the Alpine region (O'Dogherty et al., 2016). The latter taxon may be an invasive species that briefly entered the Boreal sea from the Tethyan waters, but further studies are needed to confirm this. In the Middle Oxfordian of Vorobyovy Gory, radiolarians were found in the Podmoskovye Formation (Palechek and Ustinova, 2020). The composition of the assemblage is close to the Early Oxfordian – *Praeconocaryomma* Pessagno (including *P. decora* Yeh), poorly preserved stauraxons, and a few discoid and spherical morphotypes are also present. Nassellarians are represented by parvicingulids: *Parvicingula elegans* Pessagno et Whalen and *Praeparvicingula donnae* Bragin (the latter was mistakenly identified in the paper (Palechek and Ustinova, 2020) as *Parvicingula papulata* Kozlova et Vishnevskaya, see synonymy in the article (Bragin et al., 2024)). Of interest is the presence of several species known from the Tethyan Superrealm: *Mizukidella mokaensis* O'Dogherty, Goričan et Gawlick, *Takemuraella japonica* (Takemura), *Parahsuum carpathicum* Widz et De Wever, *Pseudodictyomitrella* (?) sp. aff. *P. spinosa* Grill et Kozur. The interpretation of this distribution is still difficult due to the lack of material: this is the only record so far. Another Middle Oxfordian radiolarian assemblage is known from the Upper Volga region, in the Rybinsk area (Bragin and Kiselev, 2013). Here, in the Ioda River section, a taxonomically impoverished assemblage was found, which differs significantly from the association found in the Middle Oxfordian from the Vorobyovy Gory borehole section. It is dominated by stauraxons, represented by the genera *Paronaella* Pessagno, *Pseudocrucella* Baumgartner and *Homoeparonaella* Baumgartner. Nassellarians are rare in this assemblage and are represented only by the genus *Praeparvicingula* Pessagno, Blome et Hull. There are not enough data to explain such a difference in the composition of relatively close territorially and coeval assemblages.

The Late Oxfordian radiolarian assemblage of the Vorobyovy Gory is found in the deposits of the Kolomenskoe Series (Palechek and Ustinova, 2020). Here, the spumellarians include *Paronaella* Pessagno, *Pseudocrucella* Baumgartner, and *Orbiculiforma* Pessagno, while the nassellarians are clearly dominated by parvicingulids: *Praeparvicingula donnae* Bragin (in the paper by Palechek and Ustinova (2020) it was incorrectly identified as *Parvicingula papulata* Kozlova et Vishnevskaya, and in another case as *Parvicingula* sp. aff. *P. rothwelli* Pessagno; Bragin, 1997; Bragin et al., 2024), *P. sp. aff. P. elementaria* (Carter), *P. enormis* (Yang) (incorrectly identified by Palechek and Usti-

nova (2020) as *Parvicingula papulata* Kozlova et Vishnevskaya (Bragin et al., 2024)). Also, there are rare records of representatives of the family Echinocampidae Bragin, which may belong to the genus *Arctocapsula* Bragin.

Another record of Oxfordian radiolarians is known from the Krylatskoye area west of Moscow (Ustinova et al., 2014). Here, together with the foraminiferal assemblage of the upper part of the Upper Oxfordian (Epistomina uhligi–Lenticulina russiensis Zone), judging by the available photographs, there are *Triactoma* sp. (similar to *T. blakei* Pessagno, but incompletely preserved), as well as poorly preserved stauraxons similar to *Crucella* Pessagno, and cyrtoid nassellarians. Ustinova et al. (2014) mentioned the presence of representatives of pantanelliids and parvicingulids here, but no photographs are provided.

The richest and most representative assemblage of Oxfordian radiolarians is known from the Upper Oxfordian of the Rybaki Section (Bragin et al., 2024). Radiolarians were found in the deposits of the Upper Oxfordian Amoebocheras serratum ammonite Zone. The assemblage includes entactinarians (genus *Perispyridium* Dumitrica) and spumellarians (*Archaeocenosphaera* Pessagno et Yang, *Triactoma* Rüst, *Praeconocaryomma* Pessagno, *Crucella* Pessagno, *Orbiculiforma* Pessagno, *Archaeospongoprimum* Pessagno, *Microsandwichia* Dumitrica), which are usually represented by a small number of specimens, sometimes even single ones. The assemblage is quantitatively dominated by nassellarians, primarily parvicingulids (*Praeparvicingula blackhorsensis* (Pessagno et Whalen), *P. donnae* Bragin, *P. elementaria* (Carter), *P. enormis* (Yang)) (Pl. I, figs. 1, 2). In addition to parvicingulids, a significant role is played by the echinocampids *Echinocampe modestum* Bragin and *Nordvikella gutta* Bragin (Pl. I, fig. 7). It should be especially mentioned that no Tethyan taxa of nassellarians were found in the assemblage.

Thus, the known Oxfordian assemblages of Boreal radiolarians of Russia are taxonomically impoverished, quantitatively dominated by the genera *Praeparvicingula* and *Parvicingula*, and characterized by the absence of representatives of the family Pantanelliidae and the presence in the Upper Oxfordian of the oldest known representatives of the family Echinocampidae.

KIMMERIDGIAN STAGE

Kimmeridgian radiolarians are known from the East European Craton and the Timan-Pechora Plate. The most representative Kimmeridgian radiolarian assemblage was described from a section near the former village of Dyakovo in Kolomenskoye (Moscow), where radiolarians were found in redeposited Kimmeridgian phosphorite concretions in the basal layers of the Volgian Stage (Bragin, 1997). The spumellarian

assemblages include taxa common to the Upper Jurassic (*Archaeocenosphaera*, *Crucella*, *Paronaella*), but in addition to them, Pantanelliidae are found here, namely representatives of a group of species close to *Pantanellium meraceibaense*, which can spread into Boreal regions (Pessagno et al., 1987). This assemblage includes three species: *Pantanellium tierrablancaense* Pessagno et MacLeod, *P. huazalingoense* Pessagno et MacLeod, and *P. moscowiense* Bragin (Pl. I, figs. 8–11). The nassellarians include Parvicingulidae (*Parvicingula* sp. aff. *P. vera* Pessagno et Whalen and *Praeparvicingula donnae* Bragin), as well as representatives of the genus *Nordvikella* Bragin from the family Echinocampidae, originally assigned by the author to the genus *Pseudodictyomitrella* Grill et Kozur (Bragin, 1997).

A similar but less diverse assemblage was studied from the Cheryomukha River section (Yaroslavl Region), where it was also found in redeposited Kimmeridgian concretions at the base of the *Virgatites virgatus* ammonite Zone of the middle substage of the Volgian Stage (Bragin and Kiselev, 2013). Here, the Spumellaria includes *Archaeocenosphaera*, *Crucella*, and pantanelliids: *Pantanellium tierrablancaense* Pessagno et MacLeod and *P. meraceibaense* Pessagno et MacLeod. Nassellaria are represented by the species *Praeparvicingula donnae* Bragin; echinocampids are not recorded.

Radiolarians were also found in the Kimmeridgian part of the Gorodishchi Section (Ulyanovsk Region), but their preservation is mostly unsatisfactory, only one species is identified—*Parvicingula jonesi* Pessagno, but these forms were not illustrated (Vishnevskaya and Baraboshkin, 2001). In addition, the presence of radiolarians was noted in the lower Kimmeridgian (Upper Ermolino Formation) of the Moscow Region (Olferiev, 2012) and in the Upper Kimmeridgian (Kaluga Formation) of the Sukhinichi District of the Kaluga Region (Vishnevskaya and Mitta, 2018), but these assemblages have not yet been studied.

The “radiolarian” assemblage discovered by T.N. Palechek in the lower Kimmeridgian of the Ogarkovo Section (Kostroma region) (Palechek et al., 2021) deserves special consideration. Representatives of the subfamily Vallupinae were identified here: *Protovallupus* Pessagno et MacLeod and *Vallupus* Pessagno et Blome. This subfamily is part of the family Pantanelliidae. Vallupinae are unknown in the Panboreal Superrealm (Bragin and Bragina, 2018), moreover,

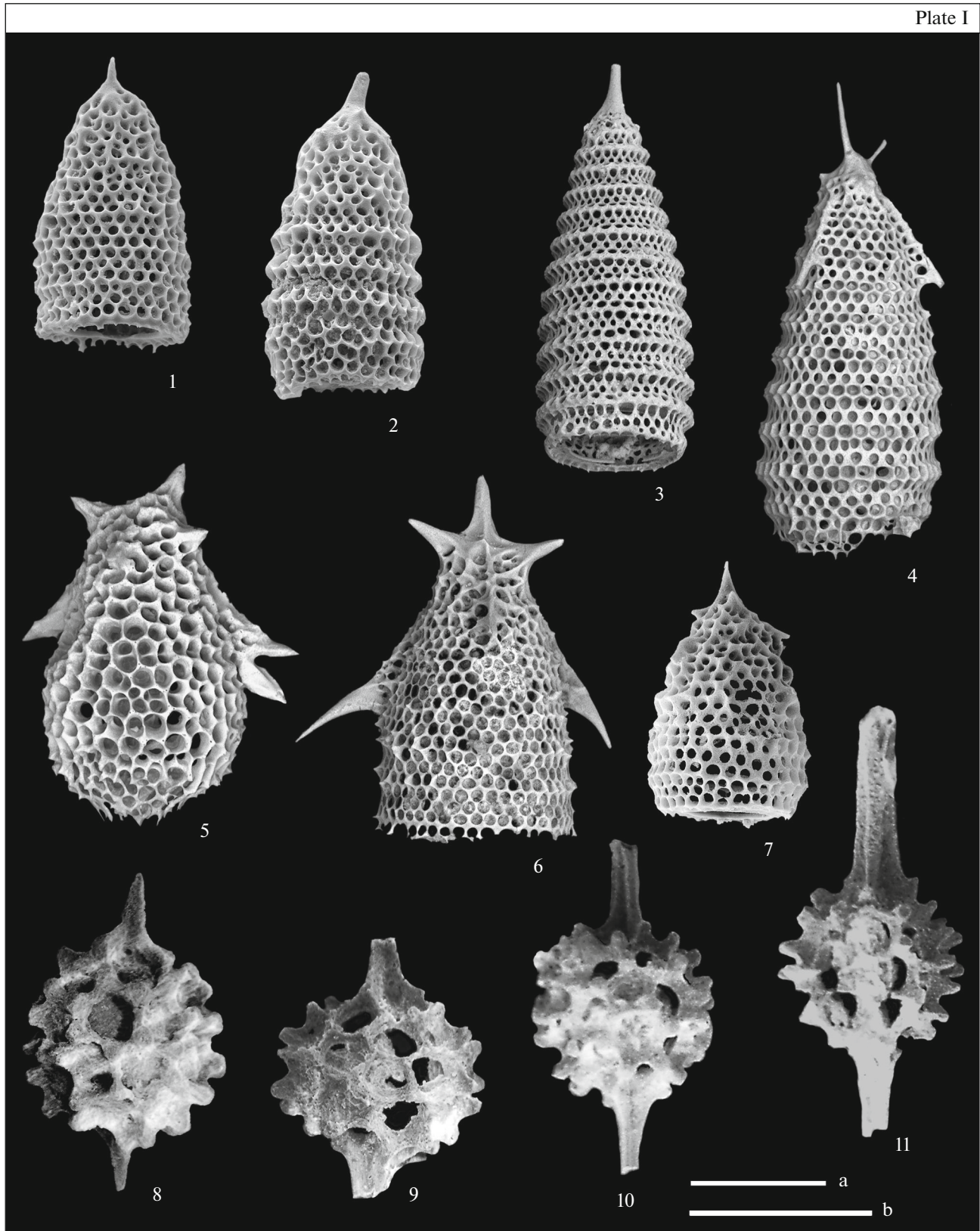
this subfamily is considered as typically circumtropical (Matsuoka, 1995), and rare finds of representatives of this subfamily in the Jurassic of the Antarctic Peninsula are interpreted as a result of the influence of a powerful meridional current that developed along the western margin of America (Kiessling, 1999). Most likely, the same explanation can be given for the Vallupinae records in Argentina (Pujana, 1995, 1996).

When examining the objects identified as Vallupinae, the following should be taken into account. These formations (see *Protovallupus* sp. in (Palechek et al., 2021, pl. 11, fig. 8) and *Vallupus* sp. in (Palechek et al., 2021, pl. 11, fig. 11)) do not have the characteristic diagnostic features of the Pantanelliidae family (largely porous latticed outer shell with hexagonal, less often pentagonal massive pore frames) and the Vallupinae subfamily (presence of the aperture and pylome). The above-mentioned morphological elements are clearly visible in the photographs of the holotypes of the type species of these genera: *Protovallupus excellens* Yang et Pessagno (Yang, Pessagno, 1989, pl. 3, figs. 4, 10) and *Vallupus hopsoni* Pessagno et Blome (Pessagno et al., 1984, pl. 1, fig. 21). In addition, the objects presented on plates (Palechek et al., 2021, pl. 11, figs. 8, 11) are two or more times as large as the corresponding radiolarians, and this cannot be explained by the replacement of siliceous material by other minerals. Pseudomorphs after radiolarians are well known: for example, pyrite after Triassic radiolarians of Kotelnny Island (Bragin, 2011), pyrite after Late Cretaceous radiolarians of Crimea (Bragina and Bragin, 2021), calcite after Triassic radiolarians of the Pamirs (Bragin et al., 2016). In all these cases, no increase in the size of the radiolarians occurred. It follows from the above that the objects from the Kimmeridgian, as well as from the Volgian and Ryazanian stages of the Ogarkovo section, where they were also encountered, are purely mineral formations: intergrowths of crystals and amorphous masses. They cannot in any way be defined as Vallupinae.

Apart from the East European Craton, radiolarian assemblages are widespread in the Kimmeridgian of the Timan-Pechora Plate. Thus, in the lower Kimmeridgian of the Pizhma River, Kozlova (1971, 1973) established an assemblage that includes spherical forms identified as *Thecosphaera* Haeckel, *Crucella* Pessagno, *Paronaella* Pessagno, discoid forms identified as *Spongodiscus* spp., and also *Pantanellium* (identified by G.E. Kozlova as *Stylosphaera lanceolata*, but

Plate I. Characteristic taxa of radiolarians of Boreal sections of the Upper Jurassic and lower Lower Cretaceous of Russia. (1–3) family Parvicingulidae: (1) *Praeparvicingula blackhorsensis* (Pessagno et Whalen), (2) *Praeparvicingula donnae* Bragin, (3) *Parvicingula khabakovi* (Zhamoida); (4–7) family Echinocampidae: (4) *Nordvikella elegans* Bragin, (5) *Arctocapsula perforata* Bragin, (6) *Echinocampe aliferum* Bragin, (7) *Echinocampe modestum* Bragin; (8–11) family Pantanelliidae: (8) *Pantanellium tierrablancaense* Pessagno et MacLeod, (9) *Pantanellium* sp. cf. *P. tierrablancaense* Pessagno et MacLeod, (10, 11) *Pantanellium moscowiense* Bragin. (1, 2, 7) Upper Oxfordian, Rybaki Section (Bragin et al., 2024, text-figs. 7A, 7F, 9D); (3, 6) the upper substage of the Volgian Stage, Nordvik Section (Bragin, 2011, pl. IV, fig. 9; pl. V, fig. 1); (4, 5) upper substage of the Volgian Stage, Nordvik Section (Bragin, 2009, pl. 2, fig. 1, fig. 3e); (8–11) Kimmeridgian, Dyakovo Section. Scale bar: 100 µm (4–6); 100 µm (figs. 1, 2, 7–11); 200 µm (3).

Plate I



this is an erroneous spelling; apparently, it should have been written as *Sphaerostylus lanceola* Parona, see (Kozlova, 1971, Fig. 1, fig. 4)).

Among the Nassellaria, Parvicingulidae are established (illustrated by G.E. Kozlova as *Eucyrtidium haeckeli* (Pantanelli)—this is probably the species *Praeparvicingula donnae* Bragin or the closely related *P. blackhorsensis* (Pessagno et Whalen)), as well as Echinocampidae (see *Stichopilium* sp.; Kozlova, 1971, Fig. 1, fig. 14). These results are confirmed by later studies (Kozlova, 1994). Similar radiolarian assemblages were also studied from the same region by V.S. Vishnevskaya (Vishnevskaya, 1998, 2001; Vishnevskaya and Pralnikova, 1999; Vishnevskaya and Murchey, 2002). They are characterized by the dominance of the typically Boreal Parvicingulidae (*Parvicingula* and *Praeparvicingula*), as well as the presence of the genus *Pantanellium*.

Thus, the known Kimmeridgian assemblages of boreal radiolarians in Russia are also characterized by taxonomic poverty and quantitative dominance of the genera *Praeparvicingula* and *Parvicingula*. At the same time, the Kimmeridgian assemblages contain the family Pantanelliidae, which is represented in a limited way—only by a group of closely related species *P. meraceibaense*, found in high-latitude regions (Pessagno et al., 1987; Bragin, 1997; Hull, 1997; Kiessling, 1999; Vennari and Pujana, 2017; Bragin and Bragina, 2018). This group includes *Pantanellium huazalingoense* Pessagno et MacLeod (Kimmeridgian–Tithonian), *P. meraceibaense* Pessagno et MacLeod (lower Oxfordian–lower Berriasian), *P. moscowiense* Bragin (Kimmeridgian), *P. quintachillaense* Pessagno et MacLeod (Oxfordian–lower Tithonian), and *P. tierrablancaense* Pessagno et MacLeod (Kimmeridgian). The appearance of Pantanelliidae may indicate an expansion of marine connections at the peak of the Kimmeridgian transgression, which resulted in increased water exchange with the southern basins. Finally, the Kimmeridgian assemblages also contain typically boreal Echinocampidae, which require taxonomic study.

VOLGIAN STAGE

Volgian Boreal radiolarians are widespread: they are known on the East European Craton and the Timan–Pechora Plate, as well as on the West Siberian Plate and along the northern framing of the Siberian Craton. The degree of their study is uneven. In the extreme north of the East European Craton, radiolarians of the Middle Volgian age (Dorsoplanites panderi ammonite Zone) are known from a borehole in the Pesha River basin (Mezen Syncline) (Lyurov and Vishnevskaya, 2000; Mitta and Vishnevskaya, 2006; Vishnevskaya and Kozlova, 2012). Various stauraxon forms are represented here, including five-rayed ones, and various representatives of the family Parvicingulidae, including *Parvicingula jonesi* Pessagno, *Praeparvicingula rotunda* Hull and many others.

In the central part of the East European Craton, despite the wide distribution of the Volgian Stage, radiolarians have been reliably identified in two areas of the Middle Volga region: in the vicinity of the cities of Ulyanovsk (Gorodishchi Section) and Syzran (Kashpir Section). In the Gorodishchi section, radiolarians were studied by V.S. Vishnevskaya (Vishnevskaya and Baraboshkin, 2001; Vishnevskaya and Murchey, 2002). In the lower substage of the Volgian stage, numerous representatives of the genus *Parvicingula* were recorded. In the middle substage, V.S. Vishnevskaya identified an assemblage with *Orbiculiforma* ex gr. *mclaughlini* Pessagno, *Stichocapsa? devorata* (Rüst), *Phormocampe favosa* Khudyaev, *Parvicingula hexagonata* (Heitzer), *P. cristata* Kozlova, *P. conica* (Khabakov), *P. aff. alata* Kozlova, *P. multipora* (Khudyaev), *P. haeckeli* (Pantanelli), *P. aff. spinosa* (Grill et Kozur), *Plathyceryphalus? pumilus* Rüst and *Lithocampe* cf. *terniseriata* Rüst. It should be noted that only one of the listed taxa is illustrated (*Parvicingula* aff. *alata*), while the remaining taxa were initially identified from thin sections in the works of Khudyaev (1931) and Khabakov (1937). Therefore, the assemblage requires additional detailed study.

In the upper substage of the Volgian stage of the Gorodishchi Section, V.S. Vishnevskaya identified the following assemblage: *Orbiculiforma* sp., *Praeconocaryomma* sp., *Parvicingula cristata* Kozlova, *P. alata* Kozlova, *P. blowi* (Pessagno), *Spinicingula* sp. and *Stichocapsa devorata* (Rüst) (Vishnevskaya and Baraboshkin, 2001). Later, the last of the taxa was described as a new subspecies — *Stichocapsa devorata arctica* Vishnevskaya et Murchey (Vishnevskaya and Murchey, 2002). Currently, this taxon belongs to the genus *Arctocapsula* (family Echinocampidae) (Vishnevskaya et al., 2014). Interestingly, in the Gorodishchi Section, this subspecies is extremely numerous. Representatives of the genus *Parvicingula* and the family Echinocampidae (N.Yu. Bragin's unpublished data) are also found in the Kashpir Section. It is noteworthy that the full composition of the late Volgian radiolarian assemblages from both locations has not yet been studied due to selective (often unsatisfactory) preservation and difficulty in extracting them from the rocks.

The Timan–Pechora Plate contains a Middle Volgian radiolarian assemblage with *Parvicingula papulata* (Fig. 3), recorded by G.E. Kozlova from a borehole in the Naryan-Mar region in the interval of the Dorsoplanites panderi and Dorsoplanites maximus ammonite zones (Kozlova, 1994). Numerous Parvicingulidae were found here, including *Praeparvicingula donnae* Bragin (see Kozlova, 1994, pl. 4, fig. 2; was identified by G.E. Kozlova as *Parvicingula haeckeli* (Pantanelli)), and Echinocampidae, later assigned to the new genus *Spinicingula* (Vishnevskaya and Kozlova, 2012) (see Kozlova, 1994, pl. 5, figs. 4, 8, 9). The assemblage, judging by the absence of images of Spumellaria in the photo plates, is not presented here in full. Another Middle Volga assemblage was estab-

System	Stage	Substage	Zones and beds based on radiolarians, age assemblages of radiolarians					
			Timan—Pechora Plate	West Siberian Plate			Vishnevskaya et al., 2020	Northern margins of the Siberian Craton
			Kozlova, 1994	Kozlova, 1983	Lipnitskaya, 2006	Amon, 2011		Bragin, 2011
Cretaceous	Ryazanian—Valanginian						Williriedellum	
	Ryazanian		Hemicryptocapsa salymica	Williriedellum salymica	Hemicryptocapsa salymicum		Parvicingula khabakovi—Williriedellum salymicum	Arctocapsula perforata
Jurassic	Volgian	Upper	?	Parvicingula rostrata	Parvicingula seria—Quasicrolanium planocephala	Parvicingula cf. rostrata—Parvicingula cf. seria	Parvicingula rotunda—Parvicingula alata	?
			Pseudocrolanium planocephala					
		Middle	?			Parvicingula cf. multipora	Parvicingula jonesi	Arctocapsula magna
			Parvicingula papulata	Parvicingula multipora				
	Kimmeridgian	Lower	?		Parvicingula multipora	Pseudodictyomitra cf. primitiva	Parvicingula antoshkinae—Parvicingula blowi	Parvicingula
			Crucella crassa					

Fig. 3. Zones and beds with radiolarians, as well as Kimmeridgian—Valanginian radiolarian assemblages of the Timan—Pechora plate, the West Siberian plate and the northern framing of the Siberian platform (using materials from the article by Vishnevskaya et al. (2020)).

lished by V.S. Vishnevskaya from boreholes in the Pechora basin and in the Barents Sea (Vishnevskaya and Pralnikova, 1999; Vishnevskaya, 2001; Vishnevskaya and Murchey, 2002); it includes many large Parvicingulidae: *Parvicingula* sp. cf. *P. obstinata* Hull, *P.* sp. cf. *P. jonesi* Pessagno, *P. elegans* Pessagno et Whalen, etc. Data on Spumellaria were not available here, possibly because of selective preservation of the assemblage.

The Late Volgian radiolarian assemblage with *Pseudocrolanium planocephala* (Fig. 3) was established by G.E. Kozlova from a borehole on Kolguev Island, from boreholes in the vicinity of Naryan-Mar, and from natural outcrops along the Izhma River in the interval of the of *Garniericeras catenulatum* and *Craspedites nodiger* ammonite zones (Kozlova, 1976, 1994). It contains abundant large nassellarians: representatives of the genus *Parvicingula*, as well as *Pyr-*

motertonium planocephalum (Kozlova). The family Echinocampidae is represented by the genus *Arctocapsula* (G.E. Kozlova identified these forms as *Anisicyrtis* sp. and *Stichocapsa devorata* Rüst; see Kozlova, 1994, pl. 7, figs. 4, 5, 7). The assemblage also includes discoid *Orbiculiforma* and spherical *Praeconocaryomma*.

On the West Siberian Plate, finds of radiolarians of the Volgian Stage are very numerous; they are confined to the oil-source Bazhenov Formation. Here, using thin sections, G.E. Kozlova recorded a Middle Volgian assemblage with *Parvicingula multipora* and the Late Volgian assemblage with *Parvicingula rostrata* (Fig. 3) (Kozlova, 1983). The thin section data do not allow a detailed characterization of the assemblages, but the presence of representatives of the genus *Parvicingula* is certain. Later, radiolarians of the Bazhenov Formation were studied by Lipnitskaya (2006); according to her data, radiolarians are also present in

the lower substage of the Volgian Stage and are represented mainly by *Parvicingula* spp., however, it is difficult to confirm these records, since these radiolarians were not illustrated in Lipnitskaya's paper. Later, radiolarians of the Volgian deposits of Western Siberia were studied by Amon (2011), whose work provides photographs of thin sections, identifies an assemblage with *Pseudodictyomitra* cf. *primitiva* (Fig. 3) in the lower part of the Middle Volgian substage, represented by poorly preserved nassellarians; it is not possible currently to positively identify them. For instance, in the specimens identified by E.O. Amon as *Pseudodictyomitra* cf. *primitiva* (Matsuoka et Yao), it is impossible to determine the nature of the porosity and it is impossible to confirm the presence of transverse ridges on the test segments (see Amon, 2011, pl. 1, fig. 3, and, for comparison, Matsuoka and Yao, 1985, pl. 1, figs. 1–5). In the upper part of the Middle Volgian substage, E.O. Amon identified an assemblage with *Parvicingula* cf. *multipora* (Fig. 3), in which, despite the very poor preservation, the presence of *Parvicingula* can be assumed (Amon, 2011, pl. 1, fig. 15). Higher, in the Upper Volgian Substage, an assemblage with *Parvicingula* cf. *rostrata*—*Parvicingula* cf. *seria* is also, unfortunately, in an unsatisfactory state of preservation.

In subsequent years, numerous new data on the Volgian radiolarians of Western Siberia were obtained thanks to successful isolation from rocks. The results were summarized by V.S. Vishnevskaya and E.O. Amon (Vishnevskaya et al., 2019a, 2019b, 2020; Amon et al., 2021, 2022; Vishnevskaya, 2021). According to their data, three radiolarian zones are recognized in the Volgian deposits of the West Siberian Plate: Lower-Middle Volgian *Parvicingula antoshkinae*—*P. blowi*, Middle Volgian *Parvicingula jonesi*, and Middle-Upper Volgian *Parvicingula rotunda*—*P. alata* (Fig. 3).

The Lower-Middle Volgian zone is distinguished in the interval of *Paravirgatites lideri* (upper part of the Lower Volgian Substage)—*Arctocrendonites anguinus* (Middle part of the Middle Volgian Substage) ammonite zones (Vishnevskaya et al., 2020; Rogov, 2021). The *Parvicingula antoshkinae*—*P. blowi* Zone contains the saturnalids *Acanthocircus Squinabol*, *Acanthocircularis* Vishnevskaya and *Spongosaturninus* Campbell et Clark (and this is the first find of saturnalids in the Boreal Jurassic of Russia) (Vishnevskaya, 2021), spherical Spumellaria *Acaeniotylopsis nordvikensis* Bragin, *Praeconocaryomma hexagona* (Rüst) (Vishnevskaya et al. (2020) erroneously reported the species *P. hexagona* as *Praeconocaryomma hexagonata*), *Tripocyclus trigonum* (Rüst), *Acastea tenuis* Hull, *Actinomma frigida* Kiessling, discoid and stauraxonic *Orbiculiforma* spp., *Angulobracchia bulbosa* Hull, *Neoparonaella delicata* Yang and *Higumastra* sp. ex gr. *H. inflata* Baumgartner.

Nassellaria are dominated by Parvicingulidae (*Parvicingula turrita* (Rüst), *P. deadhorsensis* Pessagno,

Blome and Hull, *P. antoshkinae* Vishnevskaya, *P. blowi* Pessagno), in addition to them there are representatives of the genera *Napora* Pessagno and *Saitoum* Pessagno and, what is very interesting, *Zhamoidellum ovum* Dumitrica — a species from the family Williriedellidae Dumitrica. Noteworthy here is the joint presence of species originally described from the Alpine Upper Jurassic (such as *Tripocyclus trigonum* (Rüst), probably having a wide geographical distribution), and species characteristic of Californian assemblages of mixed (Boreal-Tethyan) composition, such as *Acastea tenuis* Hull, *Angulobracchia bulbosa* Hull and *Neoparonaella delicata* Yang.

The *Parvicingula jonesi* Zone is distinguished in the upper part of the Middle Volgian Substage, within the *Laugeites groenlandicus* ammonite Zone (Vishnevskaya et al., 2020; Rogov, 2021). The zone is characterized by a very strong predominance of parvicingulids (*Parvicingula omgoniensis* Vishnevskaya, *P. blowi* Pessagno, *P. obstinata* Hull, etc.); in addition, *Napora lomoalta* Hull and *Zhamoidellum ovum* Dumitrica continue to be found, and *Triversus* sp. cf. *T. fastigatus* Hull appears (also the first find in the Boreal Jurassic). The composition of Spumellaria, judging from the publication (Vishnevskaya et al., 2020), is not rich—only the genus *Orbiculiforma*. This assemblage is clearly dominated by boreal taxa, in addition to which there are species of Boreal-Tethyan distribution.

The *Parvicingula rotunda*—*P. alata* Zone is distinguished in beds that are relatively poorly characterized by other fossils. In the lower parts of this zone, ammonites of the *Laugeites groenlandicus* and *Praechetaites exoticus* zones of the upper part of the Middle Volgian Substage are found, and above this zone, only the bivalves *Buchia* sp. cf. *B. fisheriana* (d'Orbigny) are found (Vishnevskaya et al., 2019b; Rogov, 2021), which is why the volume of the zone is not entirely clear. The assemblage of this zone is somewhat poorer than the previous ones. It does not contain a single species from the underlying beds, which requires interpretation. *Parvicingulids* (*Parvicingula alata* Kozlova et Vishnevskaya, *P. colemani* Pessagno et Blome, *Praeparvicingula rotunda* (Hull)) still dominate, and the spumellarian species *Higumastra scassoii* Kiessling is found here. At the very top of the upper substage of the Volgian Stage, Echinocampidae appear—representatives of the genus *Arctocapsula*. The assemblage has distinct Boreal features.

Another area where Volgian boreal radiolarians are known is the northern framing of the Siberian Platform. Here, in the Nordvik Section (Cape Urdyuk-Khaya, between Nordvik Bay and Anabar Gulf), radiolarians are found in the Middle Volgian and Upper Volgian deposits, with abundant ammonoids (Zakharov and Rogov, 2008; Bragin, 2009, 2011).

The Middle Volgian assemblage with *Arctocapsula magna* (Fig. 3) was found in the interval of the *Epivirgatites variabilis* ammonite Zone (upper part of the

substage without the uppermost parts) (Rogov and Zakharov, 2009; Zakharov et al., 2013). The assemblage is characterized by the presence of the spumellarians *Acaeniotylopsis nordvikensis* Bragin, *Archaeospongoprimum* sp. cf. *A. klingi* Pessagno, *Higumastratargida* Bragin, *Orbiculiforma* sp. aff. *O. teres* Hull, and *Staurosphaera* sp. cf. *S. amplissima* Foreman. The most abundant species are nassellarians of the family Parvicingulidae (*Praeparvicingula cappa* (Cortese), *P.* sp. cf. *P. sencilla* Hull) and the family Echinocampidae (*Arctocapsula magna* Bragin, *A. congelata* Bragin and *A. constantia* Bragin).

The Late Volgian-early Ryazan assemblage with *Arctocapsula perforata* (Fig. 3) from the same section is widespread in the interval of the Chetaites chetae Zone, terminal ammonite zone of the Upper Volgian Substage and the Chetaites sibiricus Zone, lowermost ammonite zone of the Ryazanian Stage (Rogov and Zakharov, 2009; Zakharov et al., 2013). This assemblage is significantly richer—it contains putative Entactinaria, *Glomeropyle polygonium* Bragin, Spumellaria *Acaeniotylopsis nordvikensis* Bragin (the same species as in the middle substage) and *Orbiculiforma* sp. aff. *O. railensis* Pessagno, as well as numerous and diverse nassellarians: family Parvicingulidae (*Parvicingula khabakovi* (Zhamoida) (Plate I, fig. 3), *Praeparvicingula rotunda* Hull), family Ultranaporidae (*Napora pyramidalis* Baumgartner) and family Tertoniidae (*Pyramotertonium planocephalum* (Kozlova)), as well as *Echinocampe aliferum* Bragin (Pl. I, fig. 6), *Nordvikkella elegans* Bragin (Plate I, fig. 4), *Arctocapsula perforata* Bragin (Pl. I, fig. 5) and many other representatives of these genera from the family Echinocampidae. The dominance of Boreal taxa is typical of the Volgian radiolarians from the Nordvik Section.

In general, the Volgian assemblages of Boreal radiolarians of Russia are characterized by a relatively impoverished taxonomic composition, the absence of Pantanelliidae, and the frequent presence of spherical, discoid, and stauraxon Spumellaria, which, however, do not become quantitatively dominant. Of great interest is the presence of representatives of Saturnaliidae in the Volgian Stage of the West Siberian Plate, which are not yet known in other sections of the Boreal Jurassic. The most important feature of the Volgian Boreal assemblages of radiolarians is the quantitative predominance and high taxonomic diversity of representatives of the genera *Parvicingula* and *Praeparvicingula*, as well as the family Echinocampidae.

Boreal radiolarians of the Lower Cretaceous (Ryazanian and Valanginian) are known from all four regions under consideration: East European Craton, Timan-Pechora Plate, West Siberian Plate and the northern framing of the Siberian Craton. They have been studied just as unevenly as the Volgian radiolarians. In the northern part of the East European Craton (the Vizinga River basin), Khudyaev (1931) identified morphotypes in thin sections of undifferentiated

Ryazan-Valanginian deposits that can be interpreted as various Parvicingulidae (Khudyaev, 1931, pl. I, figs. 30, 52), as well as putative Williriedellidae (Khudyaev, 1931, pl. I, fig. 14). Later, in the upper reaches of the Vyatka and Kama, Ryazanian-Valanginian radiolarians were studied in thin sections by Khabakov (1937). He established rich and diverse assemblages, which include morphotypes identified as Parvicingulidae (Khabakov, 1937, pl. XIII, figs. 64–66, 70) and even as *Parvicingula khabakovi* (Zhamoida), later described by A.I. Zhamoida from the Valanginian of the Koryak Upland (Dundo and Zhamoida, 1963; Khabakov, 1937, pl. XIV, figs. 82, 83, 85). In addition, among the thin sections there are sections that may belong to Echinocampidae (Khabakov, 1937, pl. XIV, fig. 91) and to Williriedellidae (Khabakov, 1937, pl. XI, figs. 21, 22, pl. XII, figs. 43, 44). At present, these localities have not yet been re-examined.

Within the Timan-Pechora Plate, radiolarians of the Ryazanian stage were studied by Kozlova (1976, 1994) from boreholes on Kolguev Island and in the Pechora-Kolvin aulacogen, as well as from outcrops along the Izhma River in the interval of the Hectoroceras kochi–Bojarkia mesezhnikowi ammonite zones. The assemblage includes discoid *Orbiculiforma*, and among the nassellarians, representatives of the family Williriedellidae. In addition, there are poorly preserved nassellarians that vaguely resemble *Arctocapsula*, but they cannot be identified (Kozlova, 1994, pl. 7, figs. 13, 14). Here G.E. Kozlova identified an assemblage with *Hemicyptocapsa salymica*, which, in addition to the Ryazanian Stage, can also be found in the upper parts of the Upper Volgian Substage (Fig. 3).

In Western Siberia, radiolarians of the Ryazanian Stage were initially identified by Kozlova (1983), who recognized an assemblage with *Williriedellum salymicum* in the interval of the Hectoroceras kochi–Bojarkia mesezhnikowi ammonite zones (Fig. 3). Subsequently, the presence of representatives of the family Williriedellidae in the Ryazanian Stage was confirmed by Lipnitskaya (2006). E.O. Amon (2011) identified an assemblage with *Parvicingula* cf. *rostrata* – *P.* cf. *seria* (Fig. 3); the presence of *Parvicingula* (Amon, 2011, pl. 1, figs. 24, 25) and Williriedellidae (Amon, 2011, pl. 1, Fig. 30) has been suggested based on photographs of thin sections, but the identification of these specimens to species or even in open nomenclature seems insufficiently substantiated. The same can be said about the identification of several forms as Echinocampidae.

Vishnevskaya et al. (2018, 2019a, 2019b, 2020) recognized the *Parvicingula khabakovi*–*Williriedellum salymicum* Zone (Fig. 3) in the upper part of the Upper Volgian Substage and in the Ryazanian Stage of the West Siberian Plate, which is characterized in the stratotype by the late Volgian ammonites *Praechetaites* sp. or late Volgian-early Ryazan *Chetaites* sp., and in other sections by ammonites of the *Surites* analogus

and *Bojarkia mезezhnikowi* zones of the Ryazanian Stage. It contains a small number of Spumellaria (*Higumastra scasso* Kiessling), a sharp predominance of Nassellaria, among which Parvicingulidae (*Parvicingula khabakovi* (Zhamoida) and other representatives of the genus *Parvicingula*), Williriedellidae (*Tricolocapsa campana* Kiessling, *Williriedellum salymicum* Kozlova and *Zhamoidellum boehmi* Kiessling) and Echinocampidae (*Nordvikella improcera* Bragin and *Arctocapsula perforata* Bragin) predominate, and finally, *Pyramotertonium planocephalum* (Kozlova) is always present. The complex of this zone has some similarities with the late Volgian-early Ryazan assemblage, described by N.Yu. Bragin from the Nordvik Section (Bragin, 2009, 2011), but at the same time has an important distinctive feature: it contains many representatives of the family Williriedellidae, not encountered in the Nordvik Section. It is also worth noting that in the Ryazanian and Valanginian stages of Western Siberia there are many species originally described by Kiessling (1999) from the Upper Jurassic–Lower Cretaceous of the Antarctic Peninsula.

Above the *Parvicingula khabakovi*–*Williriedellum salymicum* Zone in the boreholes of Western Siberia, Vishnevskaya et al. (2020) distinguished Ryazanian–Valanginian Beds with *Williriedellum*. These beds in the stratotype are characterized by ammonites of the *Hectoroceras kochi* Zone (Ryazanian Stage), and in other sections by ammonites of the *Tollia tolli* and *Neotollia klimovskiensis* zones (upper Ryazanian Stage–lower Valanginian). The assemblage here differs from the association of the previous zone in a smaller proportion of parvicingulids (*Parvicingula* sp. cf. *P. saltata* Hull), while Williriedellidae (*Williriedellum salymicum* Kozlova) and Echinocampidae (*Nordvikella improcera* Bragin) continue to dominate.

In the northern framing of the Siberian Craton, as already mentioned, radiolarians are known from the very base of the Ryazanian Stage of the Nordvik Section (Bragin, 2011). Besides, a locality in the lower reaches of the Lena River near the village of Ystannakh-Khocho (Vishnevskaya et al., 2014) contains an assemblage consisting of various Parvicingulidae in the deposits of the Ryazanian Stage. These are *Parvicingula* spp., *Praeparvicingula rotunda* Hull) and Echinocampidae (*Arctocapsula devorata arctica* Vishnevskaya et Murchey, *Spinicingula* sp. cf. *S. ceratina* Kozlova et Vishnevskaya and *Echinocampe* sp. aff. *E. aculeatum* Bragin. Spumellariae were not recorded in this assemblage, and Williriedellidae were not found. In general, this association is close to that of the upper substage of the Volgian Stage and the lower Ryazan Stage of the Nordvik Section.

This list summarizes the current extent of our knowledge of the Boreal radiolarians of the Upper Jurassic (Oxfordian–Volgian stage) and Lower Cretaceous (Ryazanian stage and Valanginian) of Russia. Nevertheless, although the datasets are incomplete

and a small number of studies are still ongoing, it is now possible to evaluate the available results, and to draw a number of conclusions and forecasts for the future.

CONCLUSIONS

The boreal radiolarian assemblages of the Upper Jurassic (Oxfordian–Volgian stage) and the Lower Cretaceous (Ryazanian stage and Valanginian) of Russia are characterized by a relatively impoverished taxonomic composition and are distinctly dominated by the order Nassellaria.

Representatives of two families of Nassellaria predominate: Parvicingulidae (*Parvicingula* and *Praeparvicingula*) and Echinocampidae (*Echinocampe*, *Nordvikella*, *Arctocapsula*, and probably *Spinicingula*), other taxa of Nassellaria are relatively rare and are not found everywhere, with the exception of the extremely common species *Pyramotertonium planocephalum* (Kozlova) in the Upper Volgian substage (the only species of the family Tertoniidae Dumitrica et Zügel found here) and Nassellaria of the family Williriedellidae, characteristic of the Ryazanian and Valanginian. Entactinaria are extremely rare, with only two records to date: *Perispyridium* in the Upper Oxfordian of the Rybaki Section and the problematic species *Glomeropyle polygonum* Bragin in the Upper Volgian substage of the Nordvik Section. Spumellaria are found everywhere, but their taxonomic composition is impoverished, and the number of specimens is low.

Three families of radiolarians are found in the Boreal Jurassic and Lower Cretaceous of Russia in certain stratigraphic intervals and may also be geographically restricted. Thus, the family Saturniidae is currently known only in the lower substage to the lower middle substage of the Volgian stage of the West Siberian Plate. The family Pantanelliidae is found only in the Kimmeridgian of the East European Craton and the Timan-Pechora Plate, and it is represented by only one genus *Pantanellium* and only one of its groups (*P. meraceibaense*). The family Williriedellidae is found in abundance in the Ryazanian and Valanginian of the West Siberian Plate and the Timan-Pechora Plate, but in other areas it is not yet known. It should be especially noted that the data on the presence of representatives of the subfamily Vallupinae in the Upper Jurassic–Lower Cretaceous deposits of the East European Craton are not confirmed.

Some assemblages of Upper Jurassic and Lower Cretaceous Boreal radiolarians of Russia contain taxa that are also found in more southern and even Tethyan regions.

For instance, the genus *Pantanellium* is common in the Kimmeridgian of European Russia. Its appearance could be associated with climate warming, but recent studies on oxygen isotopes indicate a constant water temperature in the seas of the East European Craton from the Callovian to the late Kimmeridgian (Wierz-

bowski et al., 2018). Therefore, it seems more probable that the marine connections of the Boreal basins of Russia with warm waters expanded as a result of transgression or the development of sea currents that facilitated the emergence of dispersal routes for planktonic organisms. It is noteworthy that the appearance of warm-water mollusks (ammonoids) in the Boreal sections of the Upper Jurassic–Lower Cretaceous of the East European Craton were previously interpreted in a similar way (Zakharov and Rogov, 2003; Rogov et al., 2008). A similar explanation can be offered for the appearance of the family Saturniidae in the lower and middle substages of the Volgian Stage of Western Siberia. There are also isolated cases of occurrence of Tethyan taxa, for example, the genus *Mizukidella* in the Oxfordian of the Moscow Syncline, which can be interpreted as short-term invasions.

Analysis of the stratigraphic distribution of radiolarians in the sections of the Boreal Jurassic–Lower Cretaceous of Russia shows significant changes in the composition of the assemblages. This is most clearly manifested in the sections of Western Siberia, which, like for example, the Bazhenov Formation, are characterized by facies consistency, continuity and the ubiquitous presence of radiolarians. Due to this, local zones for radiolarians have been recognized in the West Siberian boreholes. Zonal stratigraphy may be refined with further study. Note that the zones have been recognized primarily by species of the genus *Parvicingula*, which are particularly numerous and diverse in the Boreal sections.

Further research should focus on a detailed paleontological study of these assemblages, especially the characteristic Boreal taxa (*Parvicingula*, *Praeparvicingula*, Echinocampidae), clarification of the stratigraphic distribution of radiolarian species, filling in the gaps in the sequence of radiolarian assemblages (especially on the East European Craton, where there are still few data on the Volgian Stage and the Lower Cretaceous), detailing the data on taxa of southern origin and searching for interpretations of their appearance, and finally, searching for new data that could help solve the problem of the Boreal–Tethyan correlation. There are good prospects in all these areas, and this is confirmed by the increasing number of works on Upper Jurassic and Lower Cretaceous Boreal radiolarians and a clear trend to refine the data obtained.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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